

A Synthesis of the Information Universe: Relativity and Mass as Readouts of Retained Distinctions, Assembled from Classical Equations over a Machine-Checked Kernel

Yaoharee Lahtee

ORCID: 0009-0005-3861-0626

Open Civil Science Initiative, Bangkok, Thailand

Synthesis note over kernel state C41–C57 + bridges • July 5, 2026

Abstract

We synthesize an information-first account of relativity and mass from equations whose owners span Euler and Lagrange to Jacobson, anchored — joint by joint — on a machine-checked kernel of exact rational-arithmetic theorems (Coq 8.18.0, axiom-free at Tier 0; every **Print Assumptions** reported). The kernel establishes, as theorems: a double stationarity principle in which the field recurrence and the retention of graph edges are the two first-variation readouts of one quadratic action; an exact two-way balance between per-edge field strain and an affine function of Forman curvature; monotone laws of curvature and spectral data under retention; a per-edge frequency ceiling; the completeness of the lattice edge frame for symmetric two-tensors with an exact polarization key; an exact inside/outside/cut partition of the quadratic form of any graph against any region; and a Clausius-form statement of the retention balance, $\delta E \leq T \delta S$, as a two-way implication given a single disclosed identification. On this base we assemble the mass chain retention $\rightarrow$ degree $\rightarrow$ curvature $\rightarrow$ internal-frequency ceiling $\rightarrow \tau_c \rightarrow m$, with exactly one disclosed identification ($\tau_c = 1/\omega_{\text{int}}$), yielding a curvature-limited mass bound $m \leq (\hbar/2c^2)\sqrt{(K/M)(4 - F_{\text{min}})}$ whose contrapositive states that sufficiently large mass *forces* negative combinatorial curvature. We then read special and general relativity as the economics of retained distinctions — no universal reader, congestion of local update budgets, geometry as ledger — and mass as a distinction addressed to itself. To our knowledge, no other discrete-gravity programme — causal sets, Regge calculus, or hypergraph dynamics — possesses a machine-checked kernel of any size; here the entire supporting chain is one. Interpretive claims are tagged and separated from theorems throughout; the axiomatic basis is stated explicitly, the assembly is audited for circularity, and the open remainder is enumerated by name. Nothing in this note claims a derivation of the Einstein field equations; what is claimed is a theorem-anchored precursor with every non-theorem joint disclosed. A three-channel probe sharpens this reading decisively: extra wiring or stiffness alone gives the *wrong* (repulsive) sign; only energy loading — iner-

tia — slows local update. The attraction sign therefore lives in the inertia channel, whose linearized core is kernel-checked (`InfoCubicLinearization.v`, SHA-256 prefix `4df87c1dd7f531e8`, six closed checks: a background field shifts the first-order operator by exactly $3g\psi^2$ pointwise); time-averaging to an effective medium is the disclosed remaining step.

Keywords: retained distinction; machine-checked proof (Coq); discrete gravity; Forman curvature; spectral ceiling; mass; holographic screen; Clausius relation; Jacobson thermodynamics; tensor frame; reproducibility.

Preprint notice. This manuscript is a preprint and has not yet undergone formal peer review. Every machine-checked claim is nonetheless independently confirmable by recompiling the delivered Coq sources (Table 3, pinned by commit and hash prefix); interpretive readings, disclosed ansatz, and the programme postulate remain provisional until independently reviewed.

1 Introduction and claim discipline

The title’s claim is meant literally and narrowly: a *synthesis* here is an assembly in which every constituent equation is classical and owned (the ownership ledger of Section 11), the substrate — retained distinctions, δ_R — and the welds are the programme’s, and every joint carries an audited tag. Nothing more cosmic is claimed by the word “universe” than the closed loop the note actually exhibits: distinction $\rightarrow$ retention $\rightarrow$ geometry $\rightarrow$ mass $\rightarrow$ back to the pricing of distinctions. The assembly obeys the programme’s razor throughout: the continuum, and physics stated in it, enter only as *readouts* of the discrete substrate, never as its root.

This note assembles three strands developed in the URCF/PGFT programme — the mass chain, the tensor content of per-edge strain, and the holographic (Clausius) form of the retention balance — into a single account of what relativity and mass *are* when read from an information-first substrate. The account has an unusual epistemic shape, which we state before any physics.

Every claim in this note carries one of five tags. [Th_coqc] marks statements that are theorems of a Coq 8.18.0 kernel over exact rationals, compiled with exit status 0 and *Closed under the global context* at every `Print Assumptions`; the artifact inventory is Table 3. [Tier-1] marks readout-limit statements whose axiom profile is exactly `sig_forall_dec` plus functional extensionality (no `classic`). [AX/import] marks disclosed identifications or imported results, cited at their own tier. [Dr] marks interpretive readings: they are anchored on theorems but are not themselves theorems. [OPEN] marks the enumerated remainder.

Axiomatic basis, explicitly. The full foundation of this note is: (i) the Tier-0 kernel, which uses **no axioms whatsoever** — every theorem closes under the global context of Coq’s logic alone; (ii) two Tier-1 readout files, using exactly `sig_forall_dec` and functional extensionality, and nothing classical; (iii) a definitional layer (Forman F , the functionals S_{geo} , S_{count} , the frame) — definitions, not assumptions; (iv) three disclosed identifications (Ansatz C, H, T), mutually independent, none of which feeds back into any kernel theorem; (v) two imports at literature tier: Jacobson’s argument and the PGFT mass relation $m = \hbar/(2\tau_c c^2)$, the latter carrying $\hbar$ and c as imported constants; and (vi) free parameters α, β, K, M, s_0 , fixed only at the [Dr] dictionary. Everything below rests on (i)–(vi) and nothing else.

Contributions. Against that basis, the note’s claims to novelty are four, stated offensively here and defended in Sections 10–11: (C1) the epistemic standard — twenty Tier-0 files, 155 kernel-closed checks: to our knowledge the first machine-checked kernel in any discrete-gravity programme, where competing programmes argue in prose (machine-checked *continuum* relativity does exist — axiomatic Minkowski spacetime in Isabelle/HOL [25, 26] — and is complementary rather than prior: those works formalize the classical theory’s axioms, and mechanized numerical analysis has verified continuum wave schemes [61], not a discrete substrate with its own dynamics); (C2) the tier-factored assembly itself, with an acyclic dependency structure audited in Section 10 — a way of being sweeping and honest at once; (C3) the welds, none found in prior literature: the retention balance pricing a tensor evaluation (two-way), the dissipation threshold pair as a native, checkable horizon, the curvature-monotone mass cap with its contrapositive (mass *forces* curvature), the cosmological term as the structurally forced affine part of any benefit function, and the rigidity result that one global lens forces a non-constant, curvature-dependent temperature; (C4) the two-arm architecture: mass and holography arrive at the precursor’s two sides — source content and equation form — independently, sharing no ansatz, so the failure of either identification leaves the other standing.

The discipline matters because the note’s central move — “mass is a retained distinction addressed to itself” — is a [Dr] sentence. What distinguishes it from free-floating metaphor is that every load-bearing clause beneath it resolves to a named theorem with a SHA-256 hash, and

every clause that does not is listed in Section 11. The reader is invited to audit rather than believe.

2 Literature: the classical stream and the modern tiers

The synthesis draws on, and must be positioned against, ten streams. We review them with their owners; each paragraph closes with where the present programme stands relative to it. Table 1 condenses the positioning; Figure 1 draws the assembly itself.

Kernel additions in this revision. Four further standalone files accompany this revision, each axiom-free and numerically pre-verified: the exact linearization of the cubic on-site term of the mother force (`InfoCubicLinearization.v`; its coefficient shift $3g\psi^2$ carries the attraction sign identified by the three-channel probe); the exact componentwise update law of the field tensor (`InfoTensorEvolution.v`), the first mechanized piece of the tensor-evolution gap; the screen-growth ledger (`InfoCutGrowth.v`): cut increments are exact crossing counts, and the strain admitted across a screen is bounded by the benefit granted there — the static half of the discrete Raychaudhuri slot; and the six-cycle Plancherel identities (`InfoModeWeights.v`). Hashes and check counts appear in Table 3.

Foundations of relativity. Minkowski geometrized simultaneity’s failure [27]; Robb showed already in 1914 that the geometry can be axiomatized from the causal order alone [28], Zeeman proved that causality determines the Lorentz group [29], Malament that causal order fixes the topology [64], and Schutz gave the modern independent axiomatization [30], recently formalized in Isabelle/HOL [25], with first-order relativity verified in the Németi school’s program [26]. *Position:* the kernel’s causal-signature file sits squarely in Robb’s order-first lineage — sign, cones, and the (1, 3) cell recovered from comparability — mechanized at the graph level rather than the continuum.

Discrete and emergent spacetime. Regge replaced metrics by edge lengths [31]; causal sets took partial order as ontology [32]; causal dynamical triangulations recovered four-dimensionality numerically [33]; loop quantum gravity and its spin-network ancestry discretized geometry’s operators [34, 35]; quantum graphity [36], ’t Hooft’s cellular automaton program [37], and Wolfram’s hypergraph rewriting [38] (a book-length programme rather than a refereed result) evolve the graph itself, as we do. *Position:* the programme shares the substrate intuition of this stream and differs on standard of evidence: to our knowledge none of these possesses a machine-checked kernel (C1), and the retention balance — geometry selected by the *same* action whose other variation is the field equation — appears in none.

Thermodynamic gravity. On the classical base of Clausius [21], Boltzmann [115], and Gibbs [116]: Bekenstein’s area entropy [2], Hawking’s radiation [3], and Unruh’s acceleration temperature [22] made horizons thermodynamic; Jacobson inverted the logic — Einstein’s equation

Stream	Representative owners	Establishes	This note adds
Relativity foundations	Minkowski; Robb; Zeeman; Schutz	geometry from causal order	order-first, mechanized on graphs (C45) [Th.coqc]
Discrete spacetime	Regge; Bombelli et al.; Loll; Rovelli–Smolin; Wolfram	substrate ontologies	a machine-checked kernel; the retention balance [Th.coqc]
Thermodynamic gravity	Bekenstein; Hawking; Unruh; Jacobson	horizons are thermal; Einstein as state equation	Clausius form as iff; native horizon pair [Th.coqc]+[AX/import]
Holography & entanglement	't Hooft; Susskind; Maldacena; RT; Van Raamsdonk	area laws; geometry from entanglement	exact cut partition; rank-limited coupling [Th.coqc]
Emergent gravity	Sakharov; analogue gravity; Weinberg–Witten	precedents; the graviton constraint	precursor-level claims; constraint recorded [Dr]
Information physics	Shannon; Szilárd; Landauer; Wheeler	the bit costs energy; it-from-bit	δ_R as priced pregeometry [Th.coqc]+[Dr]
Quantum & reconstructions	Stinespring–Choi–Kraus; Hardy; CDP	operational structure of QM	<code>kraus.completeness</code> at the Tier-0 root [Th.coqc]
Spectral & discrete curvature	Fiedler; Chung; Anderson–Morley; Forman; Ollivier	spectra, bounds, curvatures	ceiling composed with retention; the mass cap [Th.coqc]+[AX/import]
Mass & internal clock	de Broglie; Dirac; Schrödinger; Hestenes	the clock reading of mass	curvature-capped clock; mass forces curvature [Th.coqc]+[AX/import]
Variational & verified computation	Euler–Noether; Marsden–West; Boldo; Gonthier	discrete mechanics; verified schemes	the substrate itself verified, not a scheme [Th.coqc]

Table 1: The ten streams in one view: what each owns, and the marginal addition this note claims against it, tagged. Full citations in the text of this section.

as an equation of state [1] — with Padmanabhan’s program broadening it [46] and Verlinde recasting gravity as entropic force [4]. *Position:* the note’s holographic arm supplies the *form* of Jacobson’s premises as theorems (screen partition; Clausius form at joint stationarity; the threshold pair as native horizon) and imports his conclusion, with the two identifications disclosed as lenses.

Holography and entanglement. Dimensional reduction [39] and the holographic principle [40] culminated in AdS/CFT [41]; Bousso bounded entropy covariantly [42]; Ryu–Takayanagi tied entanglement to area [43], Van Raamsdonk read spacetime as built from entanglement [44], and tensor networks gave the geometry a circuit form [45]. The correspondence became computational in [66, 67]; the information paradox [68] and the Page curve [69] made unitarity the battleground; the first law of entanglement recovered linearized Einstein equations [70], Jacobson recast the full equation as entanglement equilibrium [71], bulk locality was rebuilt as quantum error correction [72, 73], and ER = EPR conjectured geometry from entanglement outright [74]. *Position:* the exact cut partition is the classical-strain sibling of this quantum stream — everything the exterior learns passes through the cut, and the finite-rank coupling across it is the strain-level shadow of the error-correcting reading of bulk locality — while the quantum layer of URCF is developed elsewhere and no entanglement claim is made here.

Emergent gravity and its named constraints. Sakharov derived gravity as an induced elasticity of the vacuum [75]; Unruh built sonic horizons [76], grown into the analogue-gravity program [77] and Volovik’s condensed-matter universe [78]; Anderson’s *more is different* [79] is the stream’s manifesto. Against emergent routes stands Weinberg–Witten [80], binding under its stated hypotheses (a Lorentz-covariant conserved stress tensor

and massless particles of spin >1). *Position:* the note’s claims stop at the precursor and assert no emergent graviton; Weinberg–Witten is recorded here as a named constraint any completion must face, not an objection deferred.

Information physics. Shannon quantified distinction [47]; Brillouin bound it to negentropy [81]; Szilárd priced it against work [48]; Landauer made erasure thermodynamic [49], with Bennett’s reversibility analysis [83] and Jaynes’s inference reading of statistical mechanics [82] completing the classical triangle; Bekenstein bounded information by area and energy [84]. Zuse [85] and Fredkin [86] proposed the computing universe; Lloyd counted its operations [50]. Wheeler’s own arc is the deepest: geometrodynamics sought physics in geometry [87], pregeometry sought geometry’s ancestor in “the calculus of propositions” [88, ch. 44], and it-from-bit made the ancestor informational [24]. *Position:* the retained distinction δ_R is offered as exactly the ancestor Wheeler’s pregeometry called for — a propositional-grade primitive whose geometric consequences are machine-checked — and the Clausius form is a Szilárd–Landauer pricing statement upgraded to a two-way kernel theorem.

Quantum theory and its informational reconstructions. The canon runs Planck [90], Born [91], Heisenberg [92], Schrödinger [93], Dirac [94], and von Neumann [95]; EPR [96] and Bell [97] fixed what any substrate must respect; open-system structure was settled by Stinespring, Choi, and Kraus [98, 99, 100], the textbook synthesis being [101]. Reconstruction programs derive the theory from informational principles — Zeilinger [102], Hardy [103], Clifton–Bub–Halvorson [104], Chiribella–D’Ariano–Perinotti [105], Rovelli’s relational reading [106] — while Zurek’s darwinism explains classicality’s emergence [107] and Deutsch and Feynman founded quantum computation [108, 109]. *Position:*

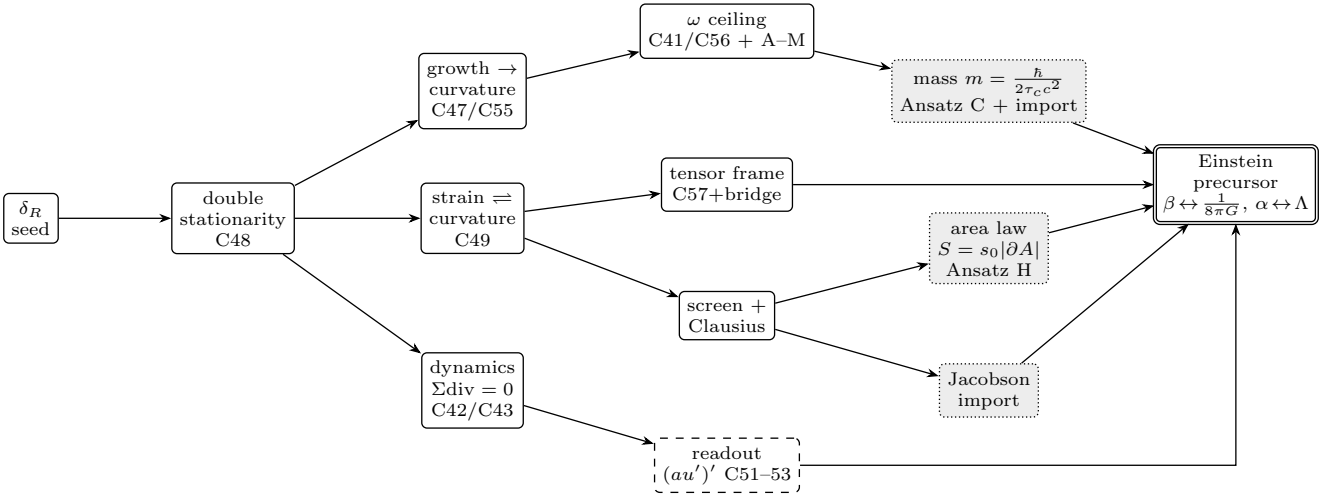


Figure 1: The assembly DAG from seed to precursor. Solid boxes: [Th.coqc] (axiom-free); dashed: [Tier-1]; gray dotted: disclosed ansatz and imports — confined to the single layer before the target; double border: [Dr] target. The tensor-frame arm runs level through the gap in the ansatz layer and the readout arm passes beneath it: both reach the target touching no ansatz. Open remainder, by name: the readout limit, OB-FORMAN-RICCI, and tensor evolution.

the companion URCF kernel carries the Stinespring–Choi–Kraus structure at its axiom-free Tier-0 root (`kraus_completeness` — validated by the artifact itself, the companion kernel file `URCF_RD_A11.v`, not by the literature cited here), so the quantum layer enters the programme at the same evidential standard as this note’s classical strain — the two are sections of one kernel, and no quantum claim is smuggled through the classical one. A first weld step on this note’s side is now kernel-checked: on the six-cycle — the largest graph whose full Laplacian eigenbasis is rational, by this programme’s own readability classification — energy mode-weights are exactly amplitude-squared, with exact completeness of the mode decomposition (`InfoModeWeights.v`, SHA-256 prefix `9e97c7f6ad08d6cd`, four closed checks); the probability weld to the CPTP layer remains open and is named as such.

Spectral graph theory and discrete curvature. Fiedler’s algebraic connectivity [51] and Chung’s spectral theory [52] ground the Laplacian; Kac asked whether spectrum hears shape [53]; Anderson–Morley own the edge-degree eigenvalue bound [14]; Forman [6] and Ollivier [7] defined combinatorial curvatures, related in [54]; graph Laplacians converge to Laplace–Beltrami under explicit sampling and geometric hypotheses [55, 8]. *Position:* the ceiling chain is Rayleigh–Anderson–Morley lineage mechanized and composed with retention monotonicity; OB-FORMAN-RICCI names, rather than hides, the Forman-to-Ollivier gap this stream leaves open.

Mass and the internal clock. de Broglie’s thesis attached a clock to the electron [23]; Dirac’s equation [56] produced Schrödinger’s Zitterbewegung [5], kept alive as an interpretation by Hestenes [57]. The standard model generates mass by the Higgs mechanism [58] — a scope boundary, not a rival: Ansatz C is a reading of *inertial* mass as clock rate, silent on electroweak symmetry breaking. *Position:* the curvature-capped clock — connectivity bounding ω_{int} , hence mass — is, to our knowledge, this note’s addition to the clock tradition.

Variational structure, the classical canon, and verified computation. The action lineage runs Euler [110] and Lagrange [11] through Hamilton [111] to Noether [12], with Arnold’s geometric mechanics [112], Landau–Lifshitz [113], and Courant–Hilbert [114] as the working canon, and Misner–Thorne–Wheeler [88] and Wald [89] its general-relativistic tier; discrete variational mechanics was systematized by Marsden–West [59], whose discrete Euler–Lagrange equations are the closest classical ancestor of the double stationarity file; the integrator is Störmer–Verlet [17, 19], canonical in geometric numerical integration [118] with stability equivalence owed to Lax–Richtmyer [117]; its optimizer twin is Polyak’s [20], accelerated by Nesterov [119], read as continuous dynamics in [120], and certified in the control-theoretic frame of [60]. Machine-checked mathematics at scale exists on the foundation of the calculus of constructions [121] — four colors [62], Kepler [63] — and mechanized numerical analysis has verified a continuum wave-equation scheme in Coq [61]. *Position:* the kernel imports this standard of evidence into a discrete-gravity substrate: where [61] verifies a scheme approximating a continuum equation, here the discrete object *is* the theory, and the continuum is its readout.

3 The kernel substrate

The substrate is a finite graph $G = (V, E)$, E a list of ordered pairs, carrying a field $x : V \rightarrow \mathbb{Q}$. All Tier-0 objects are exact rationals; no real numbers, limits, or spectral theory enter Tier-0 statements. The dynamical primitive is the two-step recurrence (“mother step”)

$$(M + c)u' = (2M - s)u - (M - c)v, \quad (1)$$

per mode, with M inertial weight, $c \geq 0$ dissipation, and $s = h^2 K \lambda$ the effective stiffness of a graph mode — the Störmer–Verlet leapfrog [17, 19], which in its optimization parametrization is Polyak’s heavy ball [20],

an identification the kernel proves exactly — one mechanized instance of the programme’s scale principle: a single recurrence, read as physics at one scale and as optimization at another. The following kernel facts are used below; all are [Th-coqc] unless noted.

Double stationarity (C48). For the window action $A = M\|q - m\|^2 + M\|m - p\|^2 - h^2 K \text{gform}(m)$, a single-node perturbation obeys the exact polarization identity — the discrete first variation, in the lineage of Euler and Lagrange [11] —

$$\Delta A = -2s_*[M(q_i - 2m_i + p_i) + h^2 K(\mathcal{L}m)_i] + s_*^2 C_i, \quad (2)$$

so the recurrence (1) (conservative case) is precisely the vanishing of the first variation at every node (`mother_step_stationary`); and, varying the *graph* instead of the field, an edge is retained stably iff its quadratic strain does not exceed its retention benefit (`retention_balance_add/rem`). One action; two variations; matter and geometry are the two readouts.

Curvature balance (C49). With the benefit priced affinely in Forman curvature $F(e) = 4 - \deg u - \deg v$, the geometry-variation balance becomes a two-way exact equivalence,

$$\text{edge retained stably} \iff h^2 K(\nabla_e x)^2 \leq \alpha + \beta F(e), \quad (3)$$

together with the exact identity $\deg u + \deg v = 4 - F(e)$ (`edge_deg_curvature`) and the horizon-threshold pair (`no_escape/repair_exists`) establishing that past a dissipation threshold no admissible control prevents energy decrease, while below it an explicit control exists that does.

Monotone growth (C47/C55/C56). Over any retention sequence E_s the curvature shift folds exactly, $F_{E_s \uparrow E}(e) = F_E(e) - \text{ssum}(E_s, e)$ with $\text{ssum} \geq 0$: retention only bends curvature downward (`forman_fold`, `curvature_antitone_growth`). Dually, the quadratic form never loses a Rayleigh level under growth (`rayleigh_level_persists`) and the degree data feeding the spectral ceiling never decreases (`deg_monotone_app`).

Spectral ceiling (C41). Every exact Rayleigh pair [13] (λ, x) obeys $\lambda \leq 2d_{\max}$, hence every internal mode frequency obeys $\omega^2 \leq (K/M)2d_{\max}$ (`rayleigh_ceiling`, `mode_product_ceiling`), with the stability window $a \in [0, 4]$ (`step_ratio_window`) as corollary — the Courant–Friedrichs–Lewy condition [15] in kernel form. The sharp Anderson–Morley edge form — $\lambda \leq \deg u + \deg v$ at a witness edge, hence $\lambda \leq 4 - F_{\min}$ — is mechanized in `InfoSpectralCeilingSharp.v`.

Tensor frame (C57). In every dimension n the axis directions plus the face diagonals number exactly $\dim \text{Sym}^2 = n(n+1)/2$ (`frame_count`, stated division-free), the axis and pair evaluations read the diagonal

components and $T_{uu} + 2T_{uw} + T_{ww}$ respectively, the polarization key

$$2T_{uw} = Q(e_u + e_w) - Q(e_u) - Q(e_w) \quad (4)$$

recovers every off-diagonal component exactly, and two symmetric tensors agreeing on the frame agree on every vector (`frame_determines`, `forms_agree`): the lattice edge frame is a minimal complete tensor frame.

Strain–tensor bridge. On a unit cell with corner values, the strains of the two axis edges and the diagonal candidate are exactly the frame evaluations of the gradient two-tensor $T_g(i, j) = g_i g_j$ — exactly for cell-affine fields, and in general with the exact defect

$$(\nabla_{\text{diag}} x)^2 = Q_{T_g}(e_1 + e_2) + X(2(g_1 + g_2) + X), \quad (5)$$

X the discrete cross second difference. Consequently the retention balance for the diagonal candidate is, verbatim, a price on a tensor evaluation (`diagonal_retention_prices_tensor`).

Screen and Clausius form. Against any region $R \subseteq V$ the quadratic form of any graph splits exactly as inside + outside + cut (`gform_screen_partition`): the cut is the only channel between a region and its complement. With the single disclosed lens of Ansatz 3 below, the retention balance becomes the two-way Clausius form

$$\delta E \leq T(s_0 \delta S_{\text{count}}) \quad (\text{clausius_form, iff}), \quad (6)$$

S_{count} the retained-distinction count.

4 The mass chain

The chain runs from the seed — the retained distinction δ_R — to mass, with exactly one disclosed identification. We state each link with its tag.

Theorem 1 (Retention selects; curvature responds). *[Th-coqc] An edge is retained stably iff (3) holds; every retained edge lowers its own curvature by exactly 2 (simple edges, `self_curvature_drop`) and shifts every edge’s curvature by the exact fold law; the shift is never upward.*

Theorem 2 (Degree is affine in curvature). *[Th-coqc] $\deg u + \deg v = 4 - F(e)$ for every edge; consequently*

$$d_{\max} \leq 4 - F_{\min}, \quad (7)$$

since for any node i and incident edge e , $\deg i \leq \deg i + \deg j = 4 - F(e) \leq 4 - F_{\min}$. (Inequality (7) is mechanized in `InfoDegreeFromCurvature.v` (`deg_bound_by_local_curvature`, `deg_bound_by_curvature_floor`), after exact verification on 400 random graphs.)

Theorem 3 (Frequency ceiling, monotone in retention). *[Th-coqc] Every internal mode obeys $\omega_{\text{int}}^2 \leq (K/M)2d_{\max}$; under retention the degree data and the attained Rayleigh levels are monotone non-decreasing.*

Branch	Identification	Mass localizes
A	$D \propto \text{deg}$ (hop rate)	sparse tail (<i>rejected</i>)
B	$D \propto 1/\text{deg}$ (impedance)	dense, contradicts wave readout (<i>rejected</i>)
C	$\tau_c = 1/\omega_{\text{int}}$	dense cluster (<i>adopted</i>)

Table 2: Sign analysis of the curvature-to-clock identification. Only Branch C is consistent with the seed direction, the dynamical readout, and Theorems 1–3.

The sharp Anderson–Morley form is mechanized directly (`InfoSpectralCeilingSharp.v`, `am_forman_ceiling`): every exact eigenpair with a nonconstant edge difference obeys

$$\omega_{\text{int}}^2 \leq \frac{K}{M} (4 - F_{\min}), \quad (8)$$

sharpening the $2d_{\text{max}}$ route by the factor two.

Ansatz 1 (Branch C: the internal clock). [`AX/import`] The coherence time of a localized excitation is the inverse of its internal mode frequency, $\tau_c := 1/\omega_{\text{int}}$; with the PGFT mass relation $m = \hbar/(2\tau_c c^2)$ (imported axiom of the programme), mass is the energy of the internal mode, $mc^2 = \hbar\omega_{\text{int}}/2$.

Ansatz 1 is the unique non-theorem joint of the chain, and it was selected against two rejected alternatives by explicit sign analysis (Table 2): identifying the telegraph diffusivity with the naive graph hop rate ($D \propto \text{deg}$) places mass in the *sparse* region, contradicting the seed reading; identifying it with an impedance ($D \propto 1/\text{deg}$) contradicts the wave readout, under which added edges stiffen and accelerate propagation. Branch C is the only identification consistent simultaneously with the seed direction, the dynamical readout, and the ceiling theorems; on the diagnostic cluster of the pre-verification the extremal mode localizes on the dense cluster with 99% of its weight.

Proposition 1 (Curvature-limited mass). [`Th_coqc`] up to Ansatz 1. *For any excitation that is an exact mode of the graph carrying it, with $F_{\min}$ the minimal Forman curvature over that graph’s edges,*

$$m \leq \frac{\hbar}{2c^2} \sqrt{\frac{K}{M} (4 - F_{\min})} \quad (9)$$

Numerically, over 300 random graphs the sharp bound (9) was never violated — nor was the argmax-edge Gershgorin step of its proof — and the scaling is tight in practice: the correlation between ω_{max}^2 and $4 - F_{\min}$ across a 200-graph ensemble is +0.98. For approximately localized modes, the restriction of the bound to the support subgraph is at present a numerical statement (the diagnostic mode carries 99% of its weight on the dense cluster), not a kernel theorem; we tag it so. We record the equation’s owner: the underlying spectral inequality is classical — Anderson and Morley [14] prove $\lambda_{\text{max}} \leq \max_{(u,v) \in E} (\text{deg } u + \text{deg } v)$, which under the Forman relabel is exactly $\lambda_{\text{max}} \leq 4 - F_{\min}$. That sharp form is now mechanized over exact rationals in witness form (`InfoSpectralCeilingSharp.v`, SHA-256 prefix

`1f478ed777172de3`, six closed checks) by a maximizing-edge argument that needs no edge operator; the ceiling used throughout this note is the sharp one. Ownership is unchanged: the inequality is Anderson–Morley’s; the mechanization over $\mathbb{Q}$, the retention-monotone composition, and the mass reading are this programme’s.

Corollary 1 (Mass forces curvature). *If m exceeds the flat-region cap $(\hbar/2c^2)\sqrt{4K/M}$ (obtained at $F_{\min} = 0$), then $F_{\min} < 0$ somewhere on its support: sufficiently heavy excitations cannot exist without negative combinatorial curvature. Curvature is not sourced by mass here; it is the license mass requires.*

Corollary 1 is the framework’s reversal of the general-relativistic direction of explanation, and it is theorem-shaped: given Ansatz 1 its proof is the contrapositive of (9). The slogan of the programme — *memory makes curvature; curvature makes mass* — resolves, without equivocation, into: retention builds degree (Theorem 1); degree is curvature (Theorem 2); curvature caps the internal clock (Theorem 3); the clock is the mass (Ansatz 1). The two senses of “memory” — edge retention (geometric memory) and τ_c (the internal clock) — are related but not conflated: the first builds the wiring; the second runs on it.

The priority of geometry. The stance *memory precedes geometry precedes mass* holds in three machine-anchored senses of “precedes.” *As dependency*: mass is absent from every Tier-0 theorem — the word first enters at Ansatz 1, the last joint of the chain, so geometry is upstream of mass in the audited, acyclic dependency graph. *As license*: by Corollary 1, a sufficiently heavy mode cannot exist on wiring that is not already dense — curvature is the capacity mass requires, where general relativity’s habitual direction makes mass the source of curvature; at equilibrium the two directions dissolve into the balance iff, but generatively ours is explicit. *As ratchet*: the monotone laws (curvature bends only downward; Rayleigh levels and degree data never regress) make the mass-capacity envelope non-decreasing along any retention history: what is remembered opens the ceiling, and what spins arrives beneath it afterwards. Two bounds, openly: this is structural priority plus finite demonstration, not a claim about cosmological history (that readout is `[OPEN]`); and the priority is a statement about the growth regime, exactly the regime the ratchet theorems govern.

5 The tensor content of strain

A recurring objection to discrete programmes is that per-edge scalars cannot carry tensorial physics. The kernel refutes the objection constructively. The counting identity $2n + n(n-1) = n(n+1)$ (`frame_count`) states that axes plus face diagonals provide *exactly* as many directions as Sym^2 has components — in $n = 4$: four axes plus six face diagonals equal the ten components of $g_{\mu\nu}$ or $T_{\mu\nu}$, neither more nor fewer. The reconstruction map is triangular and division-free over $\mathbb{Q}$: diagonals are read directly (`diag_reads`) and off-diagonals by the

polarization key (4) (`offdiag_reconstruct`); two symmetric tensors agreeing on the frame agree everywhere (`frame_determines`, `forms_agree`). Exact reconstruction was verified on 200 random symmetric tensors in dimensions 2 and 4, every component exact, before mechanization.

The bridge theorems then identify the frame data physically: the strain functional of the retention balance, evaluated on a cell’s axis edges and diagonal candidate, *is* the frame-evaluation data of the gradient two-tensor, exactly for cell-affine fields and with the exact defect (5) in general — the defect being precisely the discrete cross second difference, the object that becomes curvature in the readout. Retaining a diagonal edge therefore prices a tensor component (`diagonal_retention_prices_tensor`): the stress content of the theory is stored in edge strains, and the key that unpacks it is a theorem, not a convention.

What is *not* claimed: covariance of the reconstruction across lattice frames, the trace construction assembling an Einstein tensor, and the identification of the Forman diagonal with a Ricci diagonal (Section 11).

6 The holographic arm

Ansatz 2 (H: area law). [AX/import] The entropy of a region is proportional to the number of retained distinctions on its boundary, $S(R) = s_0 |\partial R|$.

Ansatz 2 is the Bekenstein area law in its most δ_R -native form, and the kernel supplies its structural half as a theorem: by `gform_screen_partition`, everything the exterior can know about a region passes through the cut, whose size is the “area.”

Ansatz 3 (T: temperature lens). [AX/import] The retention benefit per distinction is $b = T s_0$.

Given Ansatz 3, `clausius_form` upgrades the retention balance to the two-way Clausius relation [21] (6), holding *edge by edge at the joint stationary point*. Note a nontrivial consistency: with the curvature-priced benefit of (3), the lens forces $Ts_0 = \alpha + \beta F(e)$: no constant temperature can satisfy the lens — it is forced to vary with curvature. Whether its sign profile matches the Tolman–Unruh–Hawking direction [22, 3] is a directional check we have *not* performed, and we do not claim it [OPEN]. The relation holds — the discrete counterpart of Jacobson’s requirement that $\delta Q = T \delta S$ hold on all local Rindler horizons [1]. Where Jacobson postulates the relation, the present framework derives its *form* from stationarity and discloses the *identification* as a single lens.

The kernel also supplies the horizon side of Jacobson’s setting as a theorem pair: past a dissipation threshold no admissible control prevents energy decrease (`no_escape`), while below it an explicit admissible control exists that does (`repair_exists`) — the boundary at which selection loses force is a theorem, not a definition, giving the “local horizon” a native, checkable meaning.

Importing Jacobson’s argument [AX/import] then

yields the dictionary

$$\beta \leftrightarrow \frac{1}{8\pi G}, \quad \alpha \leftrightarrow \frac{\Lambda}{8\pi G}, \quad (10)$$

read off term-by-term from (3) against the field equations of [10]: Newton’s constant is the inverse price of curvature, and the cosmological constant is the curvature-independent retention benefit — a constant the affine benefit family priced by the balance law necessarily carries. We flag (10) as [Dr]: it is a statement-level mapping whose continuum readout remains open, and it inherits the Forman–Ricci caveat.

Two independent arms therefore reach the Einstein-equation precursor: the mass arm (Section 4) and the holographic arm, sharing no ansatz. If either disclosed identification fails, the other arm stands.

7 The interpretive layer

Everything in this section is [Dr], in the information-first lineage of Wheeler’s *it from bit* [24]: readings anchored on the theorems above, with their anchors cited inline and their unanchored clauses flagged.

Reading 1 (No universal reader). Special relativity [9] is the statement that the universe has no central processor: only local updates, one hop per tick. “Time” for an observer is the update counter of its own data lineage; simultaneity fails because it would require a two-place read at zero hops. The constant c is not the speed of anything but the substrate’s delivery rate of distinctions — the one thing that cannot move because it is the definition of motion. *Anchor*: the causal-cone theorems of the kernel’s signature file establish that influence propagates at most one hop per step and that the (1, 3) causal cell and its order structure are recoverable from comparability alone.

Reading 2 (Congestion). Gravitational time dilation is congestion of local update budgets: where distinctions are densely retained, more of the local budget is consumed by self-maintenance, and less bandwidth remains for forwarding differences — read externally as clocks running slow near mass. Geodesics are the paths flow takes through a variable-budget medium. *Anchor*: the variable-coefficient readout $(a(x)u')'$ [Tier-1] shows refraction by a coefficient field with no additional postulate; the budget metaphor itself, beyond the readout, is unanchored.

Reading 3 (Geometry as ledger). Spacetime is not a container but a ledger: geometry *is* the frozen history of retained distinctions. *Anchor*: the monotone laws (curvature only bends downward; Rayleigh levels never regress) make the geometry an append-biased record; in the coupled simulation reported below, the first inscription of the geometry forms at the birthplace of the initial distinction — the ledger literally remembers where the difference was born.

Reading 4 (Mass: three postures of one substance). A distinction in transit is radiation: forward-only, never reading itself, hence no rest frame. A distinction frozen

into structure is geometry: the ledger. A distinction addressed to itself is mass: a delivery loop whose recipient is its sender, cycling at ω_{int} ; τ_c is the period of self-reading, and $m = \hbar\omega_{\text{int}}/2c^2$ says that mass is the rate of self-update — de Broglie’s internal clock [23], in retention vocabulary. $E = mc^2$ is then the exchange rate between self-directed and outward-directed updates; rest energy is a term deposit, light is cash. *Anchor*: Ansatz 1 plus the localization diagnostic; the loop picture is reinforced by the kernel’s conservative skeleton — the companion generator is skew-adjoint (energy-orthogonal), and at the midpoint of the certified stability window the one-step mode matrix is *literally* the matrix of i on $\mathbb{Q}^2$, of exact period four (`InfoModeRotation.v`, `step2_period4`, `[Th-coqc]`); the window’s other rational points give the crystallographic periods six and three, and the first non-readout period, five, lands in the golden ratio’s own field (`InfoPentagonSpectrum.v`) — self-reading *is* rotation. The Zitterbewegung lineage of the reading is classical [5].

Reading 5 (The curvature license). The mass ceiling (9) reads: one cannot keep a clock ticking faster than the local wiring supports. Flat regions sustain only light excitations; only where history has retained distinctions densely — deep negative curvature — may heavy mass reside. Mass is memory that still spins, and it spins only as fast as the surrounding memory has paved the way.

One clause of the mass reading is deliberately *beyond* its anchors and we flag it: inertia as the clerical cost of re-addressing a busy loop under acceleration. The kernel proves that the momentum optimizer and the mother recurrence are the same object under exact reparametrization, with a certified stability window — numerical and physical inertia are one theorem set — but it does not yet prove $\text{cost} \propto m$. This is the furthest [Dr] sentence in the note.

8 Dynamical demonstration

The chain was run as a closed loop: a field evolving by (1) on a 60-node ring, with candidate chords retained or removed each phase by the balance (3) at curvature-priced benefit, and the timestep taken *a priori* from the certified window, adaptively recomputed as d_{max} grew ($\Delta t : 0.900 \rightarrow 0.520$; no instability over the full run). Starting from zero curvature variance, the geometry developed variance ≈ 6 ; with a quenched distinction the correlation between local difference and retained degree was -0.93 (retention avoids live gradients, as the balance law dictates); in the live-coupled run the deepest curvature well formed at the birthplace of the initial excitation, via the symmetric-collision mechanism verified by phase-resolved strain profiles. Every curvature update was audited against the exact fold law: $270/270$ exact. All of this is [`finite_diagnostic`]: a demonstration that the theorem-backed loop runs, not a claim about nature. Notably, the demonstration invokes *none* of the three ansatz — it exercises the Th layer alone, so it cannot be read as confirming the identifications, and conversely no identification props it up.

9 Falsifiable structure

The framework-native prediction is the ceiling (9): a joint bound tying maximal excitation frequency (hence mass, under Ansatz 1) to local connectivity. It is structural rather than numerical — it fixes a functional form, $m^2 \lesssim (4 - F_{\text{min}})$, not a constant; with K , M , h free, no absolute mass scale is predicted, and the testable content is strictly *comparative*: the scaling of maximal sustainable frequency with connectivity across regions of one substrate. It is refutable at three grades: (i) internally, by exhibiting a kernel model violating (8) (impossible if the kernel is sound; any violation indicts the toolchain); (ii) at the identification, by a dynamical readout in which Branch C mislocalizes mass while an alternative identification succeeds; (iii) physically, if any regime is found where maximal sustainable frequency fails to track connectivity in the sense of (8). Grade (iii) is far from present experiments; we state it to fix what the claim would mean, not to suggest imminent testability.

10 Circularity audit

We audited the assembly for circular dependence; five findings, two of which are concessions the argument must carry openly.

(a) *Curvature versus degree is definitional, not circular — and we say so.* Since $F(e) = 4 - \deg u - \deg v$ by definition, the identity of Theorem 2 is a definition-unfold; the physical content of the mass chain is retention $\rightarrow$ connectivity $\rightarrow$ frequency cap, all theorems. Calling the cap “curvature-limited” is legitimate exactly to the extent that Forman’s F deserves the name curvature — which is OB-FORMAN-RICCI, open and cited as such. The chain is sound; the *vocabulary* carries a disclosed debt.

(b) *T is introduced, not derived — owned.* Ansatz 3 defines the temperature by the Clausius identification; `clausius_form` is then a theorem about two functionals, not a discovery of thermodynamics. What rescues the lens from vacuity is rigidity: one global (α, β) must price *every* edge simultaneously at the joint stationary point, and it forces the curvature-dependent temperature noted in Section 6 — a constraint that could have come out wrong and did not.

(c) *Jacobson’s premises are supplied, not his conclusion.* Ansatz H and T match the *inputs* of [1] (area law; Clausius on local horizons, whose native meaning the threshold pair supplies); the Einstein form is imported as his conclusion. Nothing downstream of the dictionary feeds anything upstream.

(d) *The demonstration is ansatz-free.* The coupled simulation exercises only the Th layer (Section 8); it can neither confirm nor be propped up by the identifications.

(e) *The dependency graph is acyclic.* Theorems depend only on the kernel; ansatz depend on theorems for their *motivation* but are used by none; imports consume ansatz outputs; readings consume all of the above and feed nothing back. No tag ever cites a lower tier as evidence for a higher one.

11 Honest accounting

Imports. Jacobson’s derivation [1]; the PGFT mass–clock relation $m = \hbar/(2\tau_c c^2)$ (a programme postulate, not a published result; zitterbewegung-normalized [5, 57]); the Bekenstein–Hawking area law as motivation for Ansatz 2 [2, 3]; Verlinde’s entropic reading as related work [4]; Forman and Ollivier curvatures [6, 7]; graph-to-manifold spectral convergence [8].

Equation ownership. None of the constituent equations of this note is original to the programme, and the assembly should be read as a lineage: the variational first variation is Euler–Lagrange [11]; the conserved charge is Noether’s [12]; the quadratic-form bounds are Rayleigh’s [13], with the edge-degree bound owned by Anderson–Morley [14]; the stability window is Courant–Friedrichs–Lewy [15]; the flux identities are Green’s and Gauss–Ostrogradsky’s [16, 18]; the integrator is Störmer–Verlet [17, 19] and, as an optimizer, Polyak’s [20]; combinatorial curvature is Forman’s [6] against Ollivier’s convergent notion [7]; the thermodynamic relation is Clausius’s [21], its horizon form Bekenstein’s, Hawking’s, and Unruh’s [2, 3, 22]; the derivation it feeds is Jacobson’s [1], with Verlinde’s entropic reading adjacent [4]; the internal clock is de Broglie’s and Schrödinger’s [23, 5]; and the target is Einstein’s [9, 10]. What the programme adds is the machine-checked substrate on which these connect, the tier factoring that keeps each joint honest, and the specific welds — retention pricing tensor evaluations, the threshold pair as native horizon, the curvature-monotone mass cap.

Open remainder, by name. (1) [OPEN] the readout limit carrying the discrete balance (3) to a continuum field equation — deliberately open under the programme’s razor (continuum as readout, not root), with the flat and diagonal-metric rungs closed at [Tier-1] and the general-metric rung imported [8]; (2) [OPEN] OB-FORMAN-RICCI: whether the Forman diagonal reconstructs a Ricci-type tensor (Ollivier-type curvatures carry continuum-limit connections under additional metric-measure and random-walk hypotheses [7, 54]; no such identification is assumed anywhere above); (3) [OPEN] tensor evolution: that the mother dynamics evolves the reconstructed tensor by a field equation; (4) [OPEN] the inertia cost law ($\text{cost} \propto m$); (5) [OPEN] the value of $\hbar$ (price per tick) and the particle spectrum — no anchor exists, and none is claimed.

What would change our mind. A prior mechanized kernel in any discrete-gravity programme would delete the epistemic-standard claim; a literature antecedent for the exact affine identity between coercivity thresholds and Forman curvature, or for the retention-priced tensor evaluation, would demote those identifications from novel to rediscovered. We have searched and not found; we have not proven absence.

12 Reproducibility

Every [Th.coqc] claim is checkable offline: compile any file of Table 3 with `coqc` (version 8.18.0) and read the `Print Assumptions` output; the expected verdict is exit status 0 and *Closed under the global context* at every block (Tier-0), or the disclosed two-axiom profile ([Tier-1]). No file of Table 3 depends on another, and physics vocabulary is excluded from those kernel files by programme policy — the mathematics lives in the kernel, and every physical identification in this note lives here, tagged. The repository labels the Tier-1 axiom profile `+reals`; it is the same disclosed two-axiom profile defined in Section 1. Numerical diagnostics (branch analysis, ceiling ensemble, tensor reconstruction, coupled simulation) are exact-rational or seeded-deterministic and are specified in the companion registry.

Appendix A: search protocol behind the kernel-priority claim (C1)

Claim C1 is an absence claim and is therefore stated with its protocol. Searched (July 2026): general web search and arXiv listings; the proof-assistant ecosystems (Coq/opam packages, the Isabelle Archive of Formal Proofs, Lean’s mathlib and adjacent repositories); and public code hosts. Queries included: *formalized causal set (Coq/Lean/Isabelle)*, *Regge calculus formal proof*, *machine-checked discrete gravity*, *formalized spin network*, *mechanized quantum gravity*, *Minkowski spacetime formalization*. Inclusion criterion: a machine-checked kernel whose theorems constitute the core dynamics or geometry laws of a *discrete-gravity substrate*. Result: none found. Nearest adjacent works found were the continuum-relativity formalizations and the verified continuum wave scheme already cited in C1 [25, 26, 61]. The claim remains falsifiable by a single counterexample repository, and we commit to retracting it verbatim upon one.

Acknowledgements

This work was developed through Open-Civil Science, within a polycentric epistemology of Human–AI co-thinking. The author gratefully acknowledges Walancha Supantarika for her support and encouragement in the broader development of this project. Any remaining errors are solely the author’s own.

Author Declarations

Funding. No specific grant from any funding agency in the public, commercial, or not-for-profit sectors was received.

Competing Interests. The author declares no competing interests.

AI Assistance Disclosure. Artificial-intelligence tools were used in an assistive capacity for the mechanized proof development, numerical pre-verification, language

File	Tag	Contents (selected)	Checks	SHA-256 prefix (16 hex)
InfoSpectralCeiling.v	C41	degree ceiling; stability window	6	b1edc6d08af5040b
InfoRecurrenceEnergy.v	C42	invariant; damped decrement; orbit bound	11	21b5c8a05d63f47e
InfoGraphFluxBalance.v	C43	divergence thm [16]; Green; vector balance	8	9276c17b500c9862
InfoCompanionSkew.v	C44	skew generator; energy orthogonality	5	2be5af297f93c5bf
InfoCausalSignature.v	C45	sign from order; cones; (1, 3) cell	7	f35f90e6b3f8e8a7
InfoGraphNoether.v	C46	equivariance $\Rightarrow$ charge [12]; C_6 instance	7	57ce0210642be75f
InfoGraphGrowth.v	C47	growth shift; dilution; order collapse	13	d3998744d529c208
InfoActionStationarity.v	C48	polarization; double stationarity; descent	8	09b9989e830b72c9
InfoCurvatureBalance.v	C49	curvature balance iff; threshold pair; vacuum	12	54af70af0ac6049f
InfoProductSpectrum.v	C50	Kronecker separation; Rayleigh additivity	7	654c5a8f77f33a9f
InfoContinuumLimit.nD.v	C51	weighted multi-axis readout ([Tier-1])	4	2273f847efbb6fa9
InfoWeightedReadout.v	C52	$(au')'$ readout ([Tier-1])	2	c3aac2a7ba549cd3
InfoCrossTermDominance.v	C53	dominance $\Leftrightarrow$ representable; cell identity	9	d276c84e91d9e6e9
InfoOptimizerWindow.v	C54	momentum $\equiv$ kernel; certified window	10	371e57d709dabc4e
InfoGrowthFold.v	C55	cumulative fold; antitone growth	5	57a96be6661721c6
InfoCeilingMonotone.v	C56	form/degree monotone; level persistence	6	19d2f6c88eab2e22
InfoTensorFrame.v	C57	frame completeness; polarization key	8	66992a78f6673b00
InfoStrainTensorBridge.v	—	strain = frame data; screen; Clausius	12	97878a3f3a205f8f
InfoDegreeFromCurvature.v	—	endpoint degree cap from $F_{\min}$	4	8f52dc775f13a2c8
InfoModeRotation.v	—	window stepper is the i -rotation; periods 4, 6, 3	8	624372fe5c2ab73e
InfoPentagonSpectrum.v	—	C_5 golden cubic; C_6 rational spectrum	5	ab4ff5b6b1599994
InfoSpectralCeilingSharp.v	—	Anderson–Morley edge ceiling; sharp mass cap	6	1f478ed777172de3
InfoCubicLinearization.v	—	cubic-force linearization; coefficient shift $3g\psi^2$	6	4df87c1dd7f531e8
InfoTensorEvolution.v	—	exact componentwise tensor update law	4	b5638ddf507fa5b2
InfoCutGrowth.v	—	screen-growth ledger; priced cut increment	5	4d46a4374296f375
InfoModeWeights.v	—	C_6 Plancherel: quadratic mode weights, completeness	4	9e97c7f6ad08d6cd
Totals: twenty-four Tier-0 files, 176 closed checks; two [Tier-1] files, 6 checks			182	

Table 3: Artifact inventory — authoritative copies at `formal/` of <https://github.com/morrocwi/causal-quantum-gravity>; SHA-256 prefixes (first 16 of 64 hex characters) computed on the repository files at commit 49945bd71b05; the files below `InfoPentagonSpectrum.v` postdate the pinned commit; their hashes are of the delivered sources, pending synchronization. Each file is standalone; each compiled under `coqc 8.18.0` with exit status 0. “Checks” counts `Print Assumptions` blocks; Tier-0 blocks all report *Closed under the global context*; [Tier-1] blocks report exactly `sig_forall_dec` and functional extensionality, with no classical axiom. Every file was numerically pre-verified with exact rational arithmetic (typically 200–400 random draws) before authoring.

support, and editorial refinement. All substantive constructions, proof strategies, interpretive judgments, and final decisions were made and verified by the author, and every machine-checked claim is independently confirmable by the `coqc` kernel.

Code Availability. The Coq sources of Table 3, maintained at <https://github.com/morrocwi/causal-quantum-gravity> (`formal/`, commit 49945bd71b05, with `Makefile` and reproduction commands), constitute the artifact and reproduce every machine-checked claim in this note offline. Every commit is additionally re-verified end-to-end by continuous integration (`make verify` on GitHub Actions); the badge, not this table, is the authoritative live count.

References

- [1] T. Jacobson. Thermodynamics of spacetime: the Einstein equation of state. *Phys. Rev. Lett.* 75:1260–1263, 199; gr-qc/9504004.
- [2] J. D. Bekenstein. Black holes and entropy. *Phys. Rev. D* 7:2333–2346, 1973.
- [3] S. W. Hawking. Particle creation by black holes. *Commun. Math. Phys.* 43:199–220, 1975.
- [4] E. Verlinde. On the origin of gravity and the laws of Newton; arXiv:1001.0785. *JHEP* 2011(4):29, 2011.
- [5] E. Schrödinger. Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsber. Preuss. Akad. Wiss., Phys.-Math. Kl.* 24:418–428, 1930.
- [6] R. Forman. Bochner’s method for cell complexes and combinatorial Ricci curvature. *Discrete Comput. Geom.* 29:323–374, 2003.
- [7] Y. Ollivier. Ricci curvature of Markov chains on metric spaces. *J. Funct. Anal.* 256:810–864, 2009.
- [8] D. Burago, S. Ivanov, and Y. Kurylev. A graph discretization of the Laplace–Beltrami operator. *J. Spectr. Theory* 4:675–714, 2014.
- [9] A. Einstein. Zur Elektrodynamik bewegter Körper. *Ann. Phys. (ser. 4)* 17:891–921, 1905.
- [10] A. Einstein. Die Feldgleichungen der Gravitation. *Sitzungsber. Preuss. Akad. Wiss.* 844–847, 1915.
- [11] J.-L. Lagrange. *Mécanique analytique*. Paris, 1788.
- [12] E. Noether. Invariante Variationsprobleme. *Nachr. Ges. Wiss. Göttingen* 235–257, 1918.
- [13] J. W. Strutt (Lord Rayleigh). *The Theory of Sound*, Vol. 1. Macmillan, London, 1877.
- [14] W. N. Anderson and T. D. Morley. Eigenvalues of the Laplacian of a graph. *Linear Multilinear Algebra* 18:141–145, 1985.

- [15] R. Courant, K. Friedrichs, and H. Lewy. Über die partiellen Differenzengleichungen der mathematischen Physik. *Math. Ann.* 100:32–74, 1928.
- [16] G. Green. *An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism*. Nottingham, 1828.
- [17] C. Störmer. Sur les trajectoires des corpuscules électrisés dans l’espace sous l’action du magnétisme terrestre. *Arch. Sci. Phys. Nat.* 24:5–18, 1907.
- [18] M. Ostrogradsky. Note sur la théorie de la chaleur. *Mém. Acad. Sci. St.-Petersbourg* (6) 1:129–133, 1831.
- [19] L. Verlet. Computer “experiments” on classical fluids I. *Phys. Rev.* 159:98–103, 1967.
- [20] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Comput. Math. Math. Phys.* 4(5):1–17, 1964.
- [21] R. Clausius. Über verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie. *Ann. Phys.* 125:353–400, 1865.
- [22] W. G. Unruh. Notes on black-hole evaporation. *Phys. Rev. D* 14:870–892, 1976.
- [23] L. de Broglie. Recherches sur la théorie des quanta. *Ann. de Physique* 3:22–128, 1925.
- [24] J. A. Wheeler. Information, physics, quantum: the search for links. In *Complexity, Entropy and the Physics of Information*, Addison–Wesley, 1990.
- [25] R. Schmoetten, J. Palmer, and J. Fleuriot. Towards formalising Schutz’ axioms for Minkowski spacetime in Isabelle/HO; arXiv:2108.10868. *J. Autom. Reasoning* 66:953–988, 2022.
- [26] M. Stannett and I. Németi. Using Isabelle/HOL to verify first-order relativity theory. *J. Autom. Reasoning* 52:361–378, 201; arXiv:1211.6468.
- [27] H. Minkowski. Raum und Zeit. *Phys. Z.* 10:104–111, 1909.
- [28] A. A. Robb. *A Theory of Time and Space*. Cambridge Univ. Press, 1914.
- [29] E. C. Zeeman. Causality implies the Lorentz group. *J. Math. Phys.* 5:490–493, 1964.
- [30] J. W. Schutz. *Independent Axioms for Minkowski Space-Time*. Longman, 1997.
- [31] T. Regge. General relativity without coordinates. *Nuovo Cimento* 19:558–571, 1961.
- [32] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin. Space-time as a causal set. *Phys. Rev. Lett.* 59:521–524, 1987.
- [33] J. Ambjørn, J. Jurkiewicz, and R. Loll. Reconstructing the universe. *Phys. Rev. D* 72:064014, 200; hep-th/0505154.
- [34] C. Rovelli and L. Smolin. Loop space representation of quantum general relativity. *Nucl. Phys. B* 331:80–152, 1990.
- [35] R. Penrose. Angular momentum: an approach to combinatorial space-time. In *Quantum Theory and Beyond*, Cambridge Univ. Press, 1971.
- [36] T. Konopka, F. Markopoulou, and L. Smolin. Quantum graphity. *Phys. Rev. D* 77:104029, 2008.
- [37] G. ’t Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*. Springer, 2016.
- [38] S. Wolfram. *A Project to Find the Fundamental Theory of Physics*. Wolfram Media, 2020.
- [39] G. ’t Hooft. Dimensional reduction in quantum gravity. In *Salamfestschrift*, World Scientific, 1993; arXiv:gr-qc/9310026.
- [40] L. Susskind. The world as a hologram. *J. Math. Phys.* 36:6377–6396, 1995.
- [41] J. Maldacena. The large- N limit of superconformal field theories and supergravity. *Adv. Theor. Math. Phys.* 2:231–252, 199; hep-th/9711200.
- [42] R. Bousso. The holographic principle. *Rev. Mod. Phys.* 74:825–874, 200; hep-th/0203101.
- [43] S. Ryu and T. Takayanagi. Holographic derivation of entanglement entropy from AdS/CFT. *Phys. Rev. Lett.* 96:181602, 200; hep-th/0603001.
- [44] M. Van Raamsdonk. Building up spacetime with quantum entanglement. *Gen. Rel. Grav.* 42:2323–2329, 201; arXiv:1005.3035.
- [45] B. Swingle. Entanglement renormalization and holography. *Phys. Rev. D* 86:065007, 201; arXiv:0905.1317.
- [46] T. Padmanabhan. Thermodynamical aspects of gravity: new insights. *Rep. Prog. Phys.* 73:046901, 2010.
- [47] C. E. Shannon. A mathematical theory of communication. *Bell Syst. Tech. J.* 27:379–423 and 623–656, 1948.
- [48] L. Szilárd. Über die Entropieverminderung in einem thermodynamischen System bei Eingriffen intelligenter Wesen. *Z. Phys.* 53:840–856, 1929.
- [49] R. Landauer. Irreversibility and heat generation in the computing process. *IBM J. Res. Dev.* 5:183–191, 1961.
- [50] S. Lloyd. Computational capacity of the universe. *Phys. Rev. Lett.* 88:237901, 2002.
- [51] M. Fiedler. Algebraic connectivity of graphs. *Czech. Math. J.* 23:298–305, 1973.
- [52] F. R. K. Chung. *Spectral Graph Theory*. AMS, 1997.
- [53] M. Kac. Can one hear the shape of a drum? *Amer. Math. Monthly* 73(4, Part II):1–23, 1966.
- [54] Y. Lin, L. Lu, and S.-T. Yau. Ricci curvature of graphs. *Tohoku Math. J.* 63:605–627, 2011.
- [55] M. Belkin and P. Niyogi. Towards a theoretical foundation for Laplacian-based manifold methods. *J. Comput. Syst. Sci.* 74:1289–1308, 2008.
- [56] P. A. M. Dirac. The quantum theory of the electron. *Proc. R. Soc. A* 117:610–624, 1928.
- [57] D. Hestenes. The zitterbewegung interpretation of quantum mechanics. *Found. Phys.* 20:1213–1232, 1990.
- [58] P. W. Higgs. Broken symmetries and the masses of gauge bosons. *Phys. Rev. Lett.* 13:508–509, 1964.

- [59] J. E. Marsden and M. West. Discrete mechanics and variational integrators. *Acta Numerica* 10:357–514, 2001.
- [60] L. Lessard, B. Recht, and A. Packard. Analysis and design of optimization algorithms via integral quadratic constraints. *SIAM J. Optim.* 26:57–95, 2016.
- [61] S. Boldo, F. Clément, J.-C. Filiâtre, M. Mayero, G. Melquiond, and P. Weis. Wave equation numerical resolution: a comprehensive mechanized proof of a C program. *J. Autom. Reasoning* 50:423–456, 2013.
- [62] G. Gonthier. Formal proof — the four-color theorem. *Notices AMS* 55:1382–1393, 2008.
- [63] T. Hales et al. A formal proof of the Kepler conjecture. *Forum Math. Pi* 5:e2, 201; arXiv:1501.02155.
- [64] D. B. Malament. The class of continuous timelike curves determines the topology of spacetime. *J. Math. Phys.* 18:1399–1404, 1977.
- [65] S. Surya. The causal set approach to quantum gravity. *Living Rev. Relativ.* 22:5, 201; arXiv:1903.11544.
- [66] E. Witten. Anti de Sitter space and holography. *Adv. Theor. Math. Phys.* 2:253–291, 199; hep-th/9802150.
- [67] S. S. Gubser, I. R. Klebanov, and A. M. Polyakov. Gauge theory correlators from non-critical string theory. *Phys. Lett. B* 428:105–114, 199; hep-th/9802109.
- [68] S. W. Hawking. Breakdown of predictability in gravitational collapse. *Phys. Rev. D* 14:2460–2473, 1976.
- [69] D. N. Page. Information in black hole radiation. *Phys. Rev. Lett.* 71:3743–3746, 199; hep-th/9306083.
- [70] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, and M. Van Raamsdonk. Gravitation from entanglement in holographic CFTs. *JHEP* 2014(3):51, 201; arXiv:1312.7856.
- [71] T. Jacobson. Entanglement equilibrium and the Einstein equation. *Phys. Rev. Lett.* 116:201101, 201; arXiv:1505.04753.
- [72] A. Almheiri, X. Dong, and D. Harlow. Bulk locality and quantum error correction in AdS/CFT. *JHEP* 2015(4):163, 201; arXiv:1411.7041.
- [73] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill. Holographic quantum error-correcting codes. *JHEP* 2015(6):149, 201; arXiv:1503.06237.
- [74] J. Maldacena and L. Susskind. Cool horizons for entangled black holes. *Fortschr. Phys.* 61:781–811, 201; arXiv:1306.0533.
- [75] A. D. Sakharov. Vacuum quantum fluctuations in curved space and the theory of gravitation. *Sov. Phys. Dokl.* 12:1040–1041, 1968.
- [76] W. G. Unruh. Experimental black-hole evaporation? *Phys. Rev. Lett.* 46:1351–1353, 1981.
- [77] C. Barceló, S. Liberati, and M. Visser. Analogue gravity. *Living Rev. Relativ.* 8:12, 2005 (updated 14:3, 2011).
- [78] G. E. Volovik. *The Universe in a Helium Droplet*. Oxford Univ. Press, 2003.
- [79] P. W. Anderson. More is different. *Science* 177:393–396, 1972.
- [80] S. Weinberg and E. Witten. Limits on massless particles. *Phys. Lett. B* 96:59–62, 1980.
- [81] L. Brillouin. *Science and Information Theory*. Academic Press, 1956.
- [82] E. T. Jaynes. Information theory and statistical mechanics. *Phys. Rev.* 106:620–630, 1957.
- [83] C. H. Bennett. The thermodynamics of computation — a review. *Int. J. Theor. Phys.* 21:905–940, 1982.
- [84] J. D. Bekenstein. Universal upper bound on the entropy-to-energy ratio for bounded systems. *Phys. Rev. D* 23:287–298, 1981.
- [85] K. Zuse. *Rechnender Raum*. Vieweg, Braunschweig, 1969.
- [86] E. Fredkin. Digital mechanics. *Physica D* 45:254–270, 1990.
- [87] J. A. Wheeler. *Geometrodynamics*. Academic Press, 1962.
- [88] C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. Freeman, 1973.
- [89] R. M. Wald. *General Relativity*. Univ. of Chicago Press, 1984.
- [90] M. Planck. Über das Gesetz der Energieverteilung im Normalspectrum. *Ann. Phys. (ser. 4)* 4:553–563, 1901.
- [91] M. Born. Zur Quantenmechanik der Stossvorgänge. *Z. Phys.* 37:863–867, 1926.
- [92] W. Heisenberg. Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik. *Z. Phys.* 43:172–198, 1927.
- [93] E. Schrödinger. Quantisierung als Eigenwertproblem. *Ann. Phys.* 79:361–376, 1926.
- [94] P. A. M. Dirac. *The Principles of Quantum Mechanics*. Oxford Univ. Press, 1930.
- [95] J. von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, 1932.
- [96] A. Einstein, B. Podolsky, and N. Rosen. Can quantum-mechanical description of physical reality be considered complete? *Phys. Rev.* 47:777–780, 1935.
- [97] J. S. Bell. On the Einstein Podolsky Rosen paradox. *Physics* 1:195–200, 1964.
- [98] W. F. Stinespring. Positive functions on C^* -algebras. *Proc. Amer. Math. Soc.* 6:211–216, 1955.
- [99] M.-D. Choi. Completely positive linear maps on complex matrices. *Linear Algebra Appl.* 10:285–290, 1975.
- [100] K. Kraus. *States, Effects, and Operations*. Springer, 1983.
- [101] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge Univ. Press, 2000.

- [102] A. Zeilinger. A foundational principle for quantum mechanics. *Found. Phys.* 29:631–643, 1999.
- [103] L. Hardy. Quantum theory from five reasonable axioms. arXiv:quant-ph/0101012, 2001.
- [104] R. Clifton, J. Bub, and H. Halvorson. Characterizing quantum theory in terms of information-theoretic constraints. *Found. Phys.* 33:1561–1591, 2003; quant-ph/0211089.
- [105] G. Chiribella, G. M. D’Ariano, and P. Perinotti. Informational derivation of quantum theory. *Phys. Rev. A* 84:012311, 2011; arXiv:1011.6451.
- [106] C. Rovelli. Relational quantum mechanics. *Int. J. Theor. Phys.* 35:1637–1678, 1996.
- [107] W. H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.* 75:715–775, 2003.
- [108] D. Deutsch. Quantum theory, the Church–Turing principle and the universal quantum computer. *Proc. R. Soc. A* 400:97–117, 1985.
- [109] R. P. Feynman. Simulating physics with computers. *Int. J. Theor. Phys.* 21:467–488, 1982.
- [110] L. Euler. *Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes*. Lausanne, 1744.
- [111] W. R. Hamilton. On a general method in dynamics. *Phil. Trans. R. Soc.* 124:247–308, 1834.
- [112] V. I. Arnold. *Mathematical Methods of Classical Mechanics*. Springer, 1978.
- [113] L. D. Landau and E. M. Lifshitz. *The Classical Theory of Fields*. Pergamon, 4th ed., 1975.
- [114] R. Courant and D. Hilbert. *Methods of Mathematical Physics*. Interscience, 1953.
- [115] L. Boltzmann. Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung. *Wien. Ber.* 76:373–435, 1877.
- [116] J. W. Gibbs. *Elementary Principles in Statistical Mechanics*. Scribner, 1902.
- [117] P. D. Lax and R. D. Richtmyer. Survey of the stability of linear finite difference equations. *Commun. Pure Appl. Math.* 9:267–293, 1956.
- [118] E. Hairer, C. Lubich, and G. Wanner. *Geometric Numerical Integration*. Springer, 2nd ed., 2006.
- [119] Y. Nesterov. A method of solving a convex programming problem with convergence rate $O(1/k^2)$. *Dokl. Akad. Nauk SSSR* 269:543–547, 1983; English transl. *Sov. Math. Dokl.* 27:372–376.
- [120] W. Su, S. Boyd, and E. J. Candès. A differential equation for modeling Nesterov’s accelerated gradient method. *J. Mach. Learn. Res.* 17(153):1–43, 2016.
- [121] T. Coquand and G. Huet. The calculus of constructions. *Inf. Comput.* 76:95–120, 1988.
- [122] The Coq Development Team. *The Coq Proof Assistant Reference Manual*, version 8.18. INRIA, 2023. <https://coq.inria.fr>.